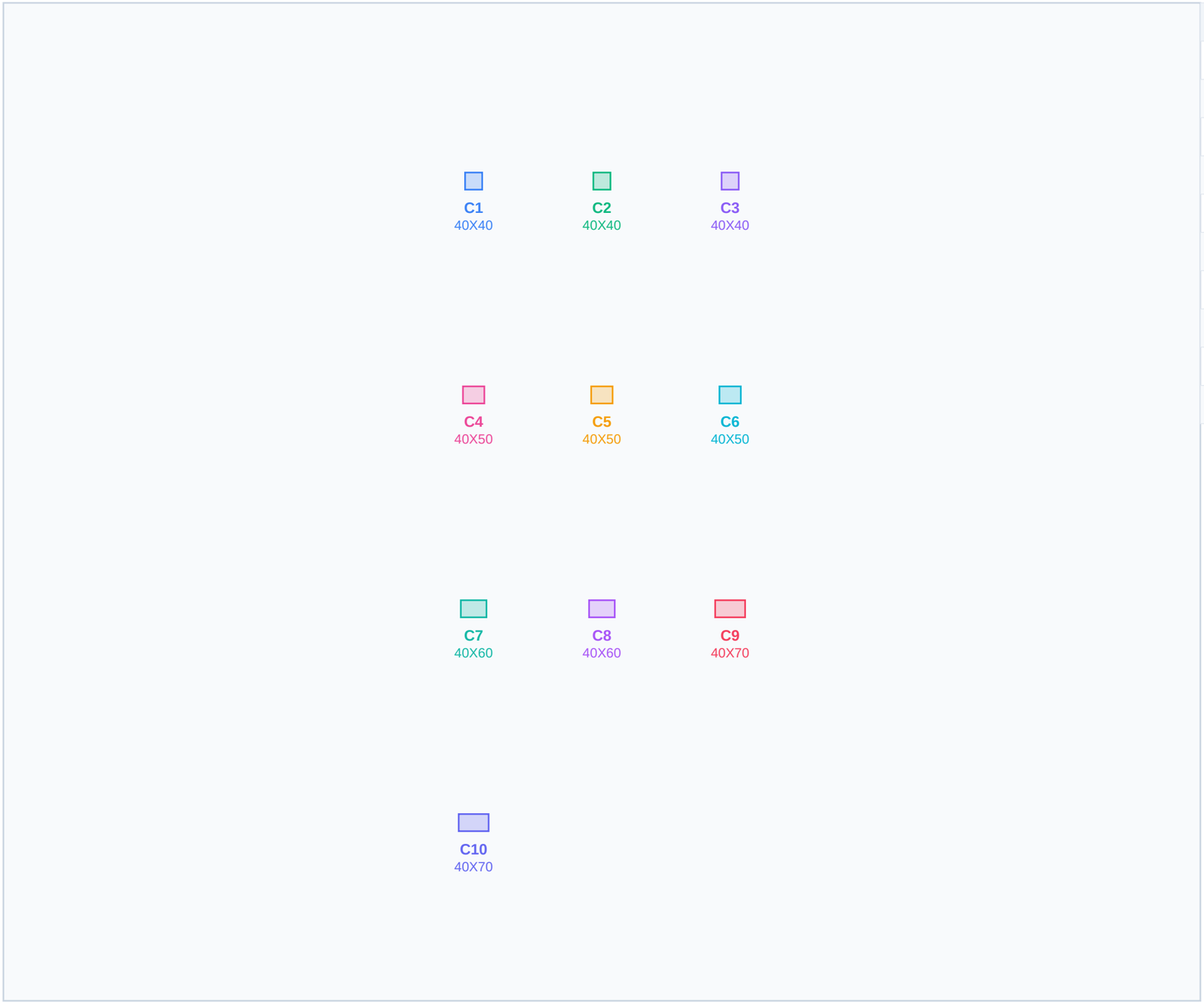
ETABS-to-CAD Column Rebar Reconciliation & Detailer Suggester

Structural Auditing and Optimization Schedule Report

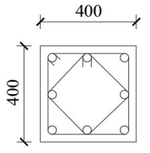
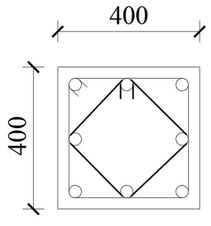
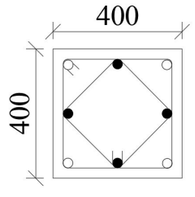
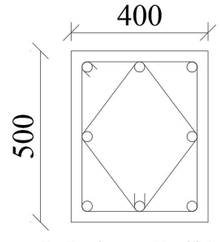
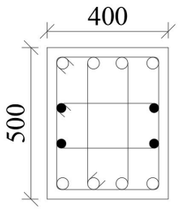
Target ETABS Model:	Mock_Etabs_Model.EDB
AutoCAD Drawing Database:	Column reinforcement.dwg (Deduplicated Schedule)
Rebar Safety Buffer:	0.00%
Total Concrete Columns Audited:	10
Conforming Matches suggested:	10 / 10 columns (100.0%)
Non-conforming (Exceeded) warnings:	0
Section Size Mismatch warnings:	0
Average Suggestion Efficiency Index:	**90.66%** (Economical Optimization)
Date of Report Generation:	2026-06-08 15:57:06

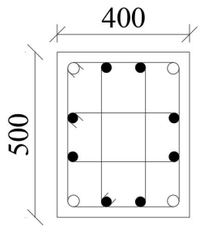
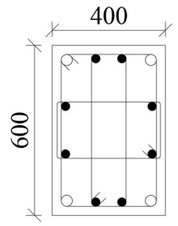
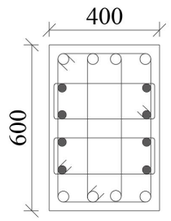
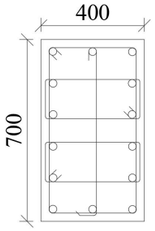
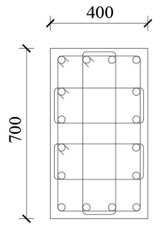
Interactive Structural Plan Layout & Groups Summary



Group	CAD Name	Size	Rebar Details	Qty
C1	C-4x4-1	400x400	Main-8 #16	1
C2	C-4x4-4	400x400	Main-8 #20	1
C3	C-4x4-7	400x400	Main - 4 #25+4 #20	1
C4	C-4x5-1	400x500	Main- 8 #16	1
C5	C-4x5-7	400x500	Main- 8 #20 +4 #16	1
C6	C-4x5-12	400x500	Main- 4 #25+8 #20	1
C7	C-4x6-4	400x600	Main- 4 #25+8 #16	1
C8	C-4x6-7	400x600	Main- 8 #20 +8 #16	1
C9	C-4x7-1	400x700	Main - 14 #16	1
C10	C-4x7-7	400x700	Main - 16 #20	1

Standard Column Rebar Schedule (AutoCAD Library)

Section	 Main-8 #16	 Main-8 #20	 Main - 4 #25+4 #20	 Main- 8 #16	 Main- 8 #20 +4 #16
Column Mark	**C1**	**C2**	**C3**	**C4**	**C5**
Suggested CAD	C-4x4-1	C-4x4-4	C-4x4-7	C-4x5-1	C-4x5-7
T1 (Ties Support)	T1	T1	T1	T1	T1
T2 (Ties Span)	T2	T2	T2	T2	T2
H1 (Confinement Ht)	H1	H1	H1	H1	H1

Section	 Main- 4 #25+8 #20	 Main- 4 #25+8 #16	 Main- 8 #20 +8 #16	 Main - 14 #16	 Main - 16 #20
Column Mark	**C6**	**C7**	**C8**	**C9**	**C10**
Suggested CAD	C-4x5-12	C-4x6-4	C-4x6-7	C-4x7-1	C-4x7-7
T1 (Ties Support)	T1	T1	T1	T1	T1
T2 (Ties Span)	T2	T2	T2	T2	T2
H1 (Confinement Ht)	H1	H1	H1	H1	H1

Structural Optimization & Audit Executive Summary

This report reconciles the required reinforcement percentages computed by CSI ETABS concrete design with the standard column detailed options available inside the AutoCAD drawing schedule. The suggested columns represent the most cost-effective and steel-economical conforming choices, checking dimensions in both orientation angles. Columns marked with **REINFORCEMENT EXCEEDED** require either cross-section size expansions or custom detailed rebar schedules, as their required reinforcement exceeds the drawing limits.

Total Columns	Conforming	Exceeded	Average Efficiency
10	**10**	0	**90.7%**

Column Mapping & Detail Selection Schedule

Notation	Story Range	Column Label	Grid	ETABS Size	Req %	CAD Name	CAD %	Rebar Details	Eff %
C1	Story 1 to Story 5	C1	A-1	400x400	0.90%	**C-4x4-1**	1.00%	Main-8 #16	**90.0%**
C2	Story 1 to Story 5	C2	A-2	400x400	1.40%	**C-4x4-4**	1.57%	Main-8 #20	**89.2%**
C3	Story 1 to Story 5	C3	A-3	400x400	1.80%	**C-4x4-7**	2.00%	Main - 4 #25+4 #20	**90.0%**
C4	Story 1 to Story 5	C4	B-1	400x500	0.70%	**C-4x5-1**	0.80%	Main- 8 #16	**87.5%**
C5	Story 1 to Story 5	C5	B-2	400x500	1.50%	**C-4x5-7**	1.66%	Main- 8 #20 +4 #16	**90.4%**
C6	Story 1 to Story 5	C6	B-3	400x500	2.10%	**C-4x5-12**	2.24%	Main- 4 #25+8 #20	**93.7%**
C7	Story 1 to Story 5	C7	C-1	400x600	1.30%	**C-4x6-4**	1.49%	Main- 4 #25+8 #16	**87.2%**
C8	Story 1 to Story 5	C8	C-2	400x600	1.60%	**C-4x6-7**	1.71%	Main- 8 #20 +8 #16	**93.6%**
C9	Story 1 to Story 5	C9	C-3	400x700	0.90%	**C-4x7-1**	1.00%	Main - 14 #16	**90.0%**
C10	Story 1 to Story 5	C10	D-1	400x700	1.70%	**C-4x7-7**	1.79%	Main - 16 #20	**95.0%**